

Control	└ active : Momentum exchange
	(Momentum Wheel, Control Momentum Gyro),
	Magnetic torque
	└ passive: Spin stabilization, Dual spin stabilization, Gravity gradient stabilization

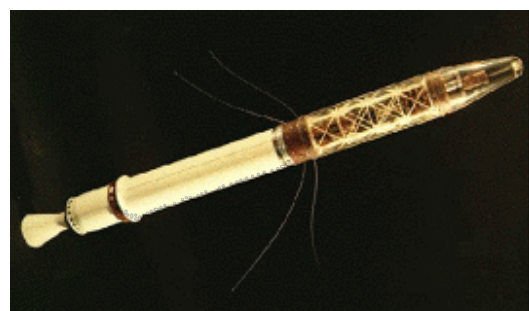
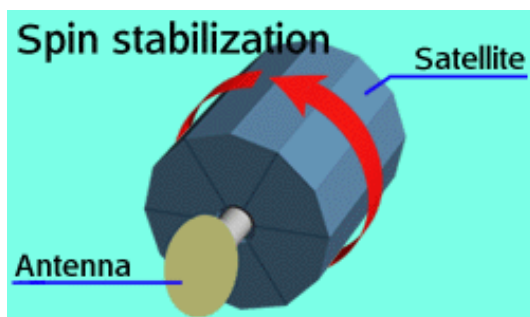
Chapter 2. Spin Stabilization

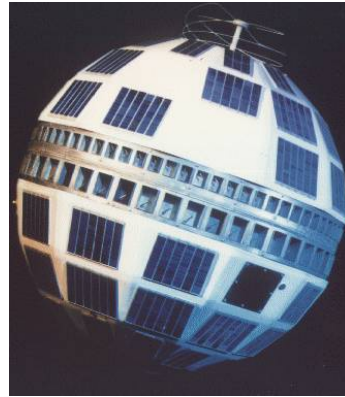
A spin stabilized spacecraft is one in which the entire spacecraft spins around one of the principal axes. The spacecraft is highly resistant to external disturbing torque because of the comparatively large angular momentum contained in the spin.

Advantages: spin stabilization is useful in that requires simplicity low cost, modest pointing accuracy and minimal maneuvering.

Disadvantages: pointing accuracy is low (0.3~1 deg).
 Tight control of the moment of inertia is required
 The only location for solar energy is the spinning body interior.
 The large angle reorientation is limited.
 Body pointing of payload sensors and antenna is not possible.

Example: Explorer I , Intelsat I , II , III , Telstar, Syncom, Meteosat, Cluster





The basic control requirements

- i) Deployment and spin-up
- ii) Sun/Earth acquisition
- iii) Spin and nutation control
- iv) Attitude pointing control

Response of the Uncontrolled vehicle

The spinning satellite can be approximated by a spinning cylinder.

$$L = \begin{bmatrix} A & 0 & 0 \\ 0 & A & 0 \\ 0 & 0 & C \end{bmatrix} \quad C : \text{spin axis moment of inertia}$$

The satellite is spin up by spin thrusters after separation from the launch vehicle.

$$\vec{M} = \frac{d\vec{H}}{dt}$$

Torque $\vec{M} = \vec{R} \times \vec{F}$,

R: thrust moment arm about the spin axis

Pure spin $RF = C \frac{d\omega}{dt}$; external torque by thrust F

위성의 자세 유지를 위한 각속도의 변화량이 주어질 때
 R: 위성 제작 시 결정된 값
 F: Thruster 제작 시 결정된 값
 ↓
 Δt를 구한다
 ⇒ I 를 구하고 정해진 연료의 I_{sp}를 이용하여 필요한 연료의 양을 알 수 있다.

$$\rightarrow \Delta t = \frac{C}{FR} \Delta \omega$$

spin up impulse $I = F \Delta t$

$$\text{spin up propellant weight } W = \frac{I}{I_{sp}}$$

$$\rightarrow I_{sp} = \frac{I}{W} = \frac{I}{m_p g_0}$$

$$\rightarrow m_p = \frac{I}{I_{sp} g_0}$$

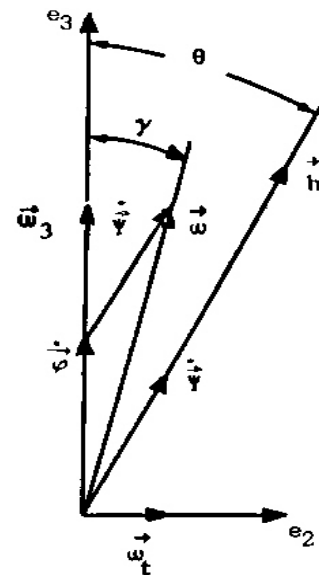
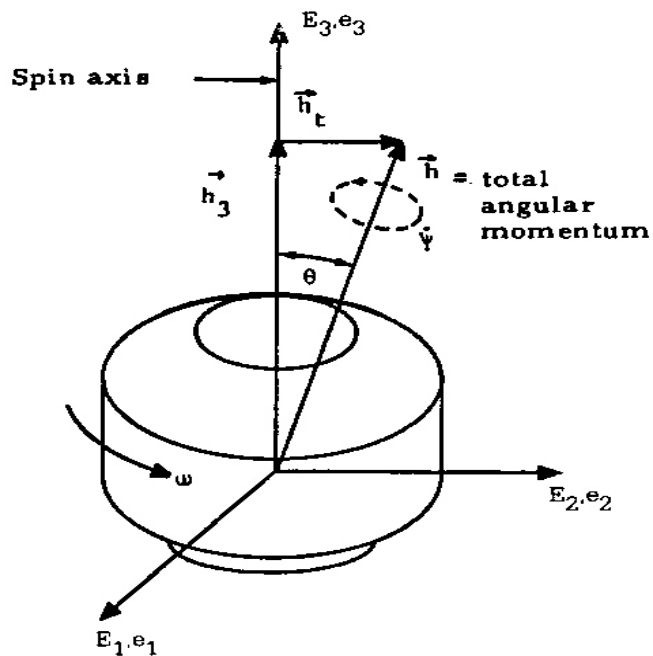
g_0 : gravitational acceleration, I_{sp} : specific impulse, m_p : propellant mass

Imperfect operation of thrust

⇒ There exist angular momentum of the transverse axis (h_t)

⇒ The spin axis is coning (precession) about a new angular momentum vector.

⇒ not pure spin



θ : nutation angle

Precession of the satellite

$$\vec{\omega} = \dot{\phi} + \dot{\psi},$$

$\dot{\phi}$ spin velocity along b_3 ,

$\dot{\psi}$ precession velocity due to h_t

$$\vec{H} = \begin{bmatrix} h_1 \\ h_2 \\ h_3 \end{bmatrix} = \begin{bmatrix} A & 0 & 0 \\ 0 & A & 0 \\ 0 & 0 & C \end{bmatrix} \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix} \quad \rightarrow h_1 = A\omega_1, \quad h_2 = A\omega_2, \quad h_3 = C\omega_3$$

$$h_t = \sqrt{h_1^2 + h_2^2} = \sqrt{A^2\omega_1^2 + A^2\omega_2^2} = A\sqrt{\omega_1^2 + \omega_2^2} = A\omega_t$$

$$h_t = A\omega_t = A\omega \sin \gamma$$

$$h_3 = C\omega_3 = C\omega \cos \gamma$$

$$\tan \theta = \frac{h_t}{h_3} = \frac{A\omega \sin \gamma}{C\omega \cos \gamma} = \frac{A}{C} \tan \gamma$$

$$h_3 = h \cos \theta = C\omega_3 = C(\dot{\phi} + \dot{\psi} \cos \theta)$$

From the previous note,

$$\dot{\psi} = \frac{C\dot{\phi}}{(A-C)\cos \theta}$$

$$\rightarrow \dot{\phi} = \frac{\dot{\psi}(A-C)}{C} \cos \theta$$

$$\therefore h_3 = C \left(\frac{A-C}{C} + 1 \right) \dot{\psi} \cos \theta = A\dot{\psi} \cos \theta$$

or

$$\begin{aligned} \dot{\psi} &= \frac{h_3}{A \cos \theta} = \frac{h}{A} \\ &\approx \frac{h_3}{A} \quad \text{for } \theta \ll 1 \quad (\text{small } \theta \text{ angle}) \end{aligned} \quad (2.10)$$

For $C > A$, retrograde precession occurs because $\dot{\psi}$ is negative.

$$\text{Precession frequency} : \frac{C-A}{A} \omega_3$$

Stability of motion

The kinetic energy

$$\begin{aligned}
 E_k &= \frac{1}{2} \vec{h} \cdot \vec{\omega} \\
 &= \frac{1}{2} (h_3 \hat{b}_3 + h_t \hat{b}_2) \cdot (\omega_3 \hat{b}_3 + \omega_t \hat{b}_2) \\
 &= \frac{1}{2} (h_3 \omega_3 + h_t \omega_t) \\
 &= \frac{1}{2} (h \cos \theta \omega \cos \gamma + h \sin \theta \omega \sin \gamma) \\
 &= \frac{1}{2} h \omega (\cos \theta \cos \gamma + \sin \theta \sin \gamma)
 \end{aligned}$$

$$\text{where } \sin \gamma = \frac{\omega_t}{\omega}$$

$$\cos \gamma = \frac{\omega_3}{\omega} = \frac{h_3}{C \omega}$$

$$\cos \theta = \frac{h_3}{h}$$

$$\begin{aligned}
 E_k &= \frac{1}{2} h \omega \left(\frac{h_3}{h} \frac{h_3}{C \omega} + \frac{\omega_t}{\omega} \sin \theta \right) \\
 h_3 &= h \cos \theta, \quad \omega_t = \frac{h_t}{A} = \frac{h \sin \theta}{A}
 \end{aligned}$$

$$\begin{aligned}
 E_k &= \frac{1}{2} h \left(\frac{h}{C} \cos^2 \theta + \frac{h}{A} \sin^2 \theta \right) \\
 &= \frac{1}{2} \frac{h^2}{C} \left(1 - \sin^2 \theta + \frac{C}{A} \sin^2 \theta \right) \\
 &= \frac{h^2}{2C} \left[1 + \left(\frac{C-A}{A} \right) \sin^2 \theta \right]
 \end{aligned}$$

Differentiating with respect to time

$$\dot{E}_k = \frac{dE_k}{dt} = \frac{h^2}{C} \left(\frac{C-A}{A} \right) \sin \theta \cos \theta \dot{\theta} \quad (2.14)$$

(일반적으로 위성(semi-rigid body) 내에서는 에너지가 감소한다.)

For $\frac{dE_k}{dt} < 0$

If $C-A > 0$ (oblate)

$\rightarrow \dot{\theta} < 0$, (precession angle decreases)

i.e., stable!!

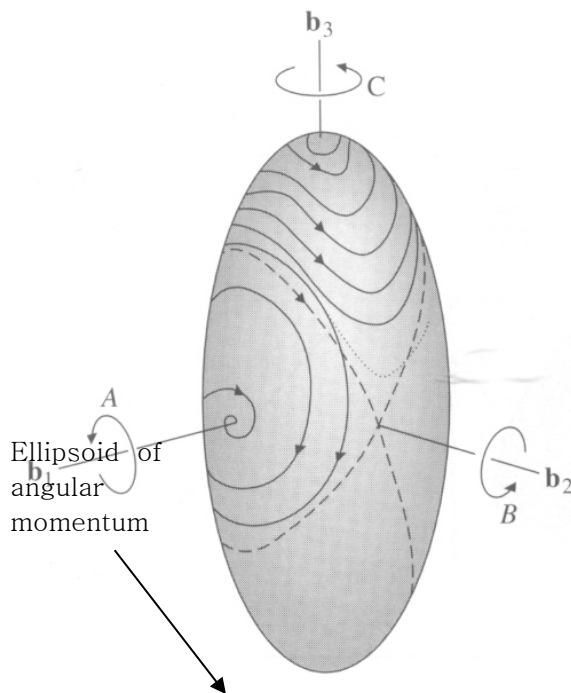
If $A-C > 0$ (prolate)

$\rightarrow \dot{\theta} > 0$,

(precession angle increases until the body spins about its transverse axis(major axis))

i.e., unstable!!

Internal motions of spacecraft components are totally unable to alter the angular momentum of spacecraft, since internal torques always cancel. However internal motions may be able to convert kinetic energy of rotation into heat by energy dissipation, in which case the energy is lost forever.



The semi-rigid body, conserving angular momentum but slowly dissipating energy is important in spacecraft dynamics. Where energy is lost to the rotational motion, the $\vec{\omega}$ vector is moving towards lower energy levels.

Figure: polhodes (trace of angular velocity on the same angular momentum ellipsoid) for semi-rigid body ($A > B > C$). Initially the body rotates about $\hat{b}_3$ axis $\rightarrow$ After energy dissipation, the body will be rotate about the major axis ($\hat{b}_1$).

$$h^2 = A^2 \omega_1^2 + B^2 \omega_2^2 + C^2 \omega_3^2, \quad \frac{\omega_1^2}{\left(\frac{h}{A}\right)^2} + \frac{\omega_2^2}{\left(\frac{h}{B}\right)^2} + \frac{\omega_3^2}{\left(\frac{h}{C}\right)^2} = 1$$

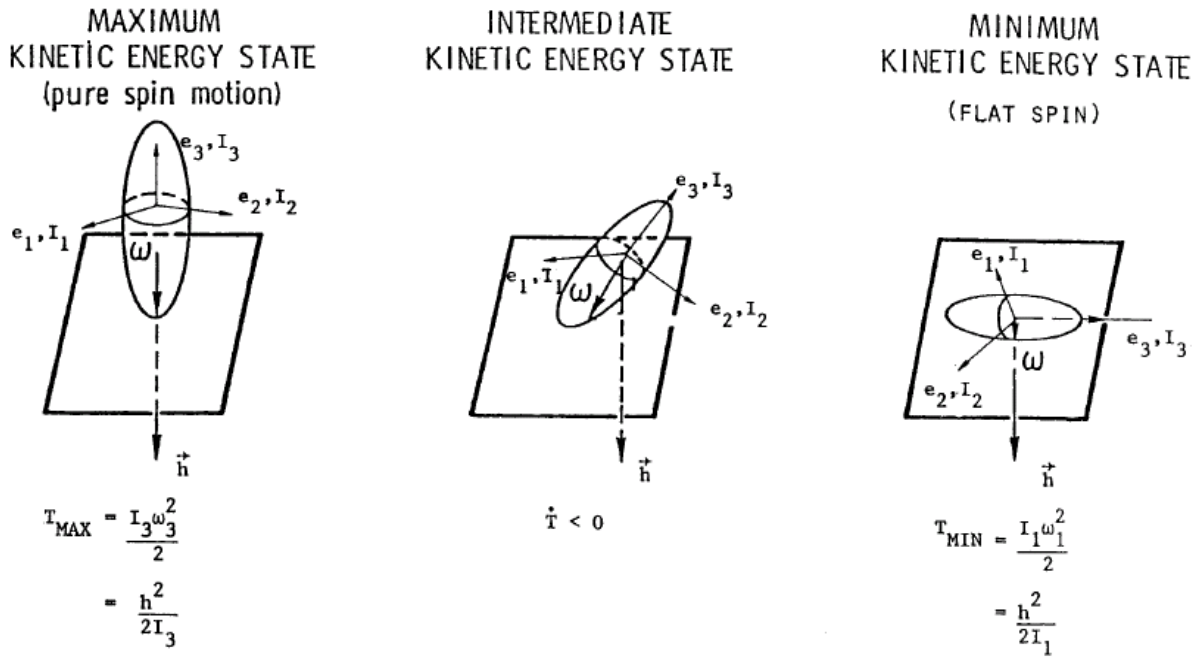
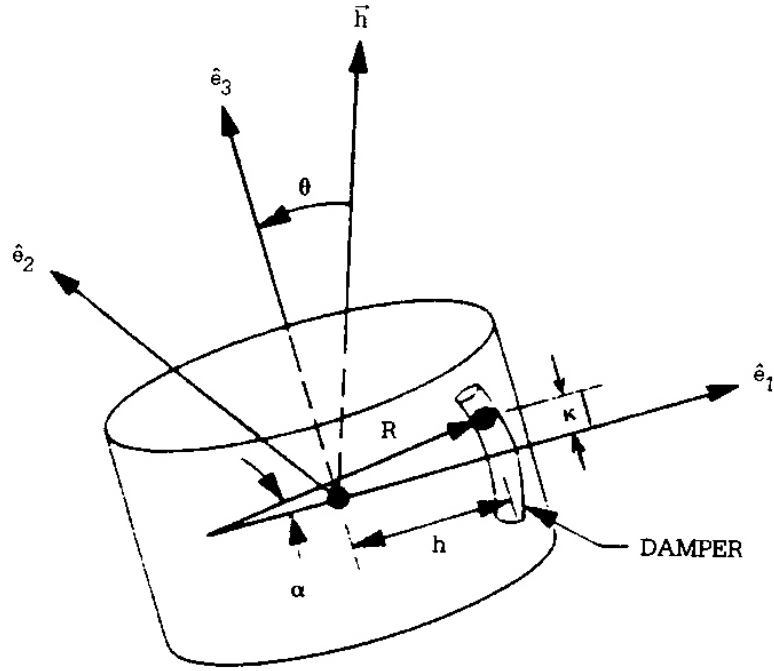


Figure 2.4 Effects of energy dissipation on the motion of a prolate spheroid.

Passive Nutation Control

The Explorer I was a prolate axisymmetric spacecraft, designed to spin about its symmetry axis for stability during the boost to orbit. Once in orbit, four whip antennas were deployed. Within only a few orbits, the energy dissipated in the flexing of these antennas converted the motion of the satellite into a stable state of tumbling about the major axis.

An oblate spinning spacecraft can be made more stable by including a “nutation damper”, a deliberate attempt to increase energy dissipation. A nutation damper would actually drive a prolate spacecraft unstable all the faster. However, most satellite designers have preferred to maintain the option of launching on expendable booster (prolate payload shroud), so prolate satellites are still common.



For spacecraft spinning about the major axis, the nutation angle θ can be reduced to zero by a device (damper). One type of damper is a ball-in-tube system. The ball moves in a tube in response to the nutation acceleration, and the energy dissipation takes place.

The magnitude of the viscous damping force is

$$f = k\dot{x}$$

where k is damping coefficient, dx/dt is instantaneous velocity of ball.

The differential work done by the ball in a viscous fluid on the satellite.

$$\begin{aligned} dW &= \vec{F} \cdot d\vec{r} \\ &= -k\dot{x} dx \quad \text{where } dx \text{ is differential displacement of the ball from equilibrium.} \\ &= -k\dot{x} \frac{dx}{dt} dt \\ &= -k\dot{x}^2 dt \end{aligned}$$

The average work per cycle can be obtained by integrating;

$$W_{av} = -\frac{k}{P} \int_0^P \dot{x}^2 dt$$

P; period of damper motion

The average work per cycle is equal to the rate of change of E_k

$$\begin{aligned} W_{av} &= \dot{E}_k \\ &= \frac{h^2}{C} \left(\frac{C-A}{A} \right) \dot{\theta} \sin \theta \cos \theta \\ &\approx \frac{h^2}{C} \left(\frac{C-A}{A} \right) \theta \dot{\theta} \quad \text{for } \theta \ll 1 \end{aligned} \quad (2.19)$$

Assume : $W_{av} = -\theta^2 g(\alpha_1, \alpha_2, \alpha_3, \dots)$

g represents a function of the system parameters.

Equation (2.19) becomes

$$\frac{h^2}{C} \left(\frac{C-A}{A} \right) \theta \dot{\theta} + \theta^2 g = 0$$

$$\dot{\theta} + \frac{\theta}{\tau} = 0$$

$$\text{where } \tau = \frac{h^2}{gC} \left(\frac{C}{A} - 1 \right)$$

The solution is

$$\therefore \theta = \theta_0 e^{-t/\tau}$$

where θ_0 ; initial nutation angle

If the initial angular momentum is given by

$$\begin{aligned} h_0 &= [h_3^2 + h_t^2]^{1/2} \\ &= [(C\omega_3)^2 + (A\omega_t)^2]^{1/2} \quad \text{when } t = t_0 \end{aligned}$$

And then,

$$h_f = C\omega_f \quad \text{when } t = t_f$$

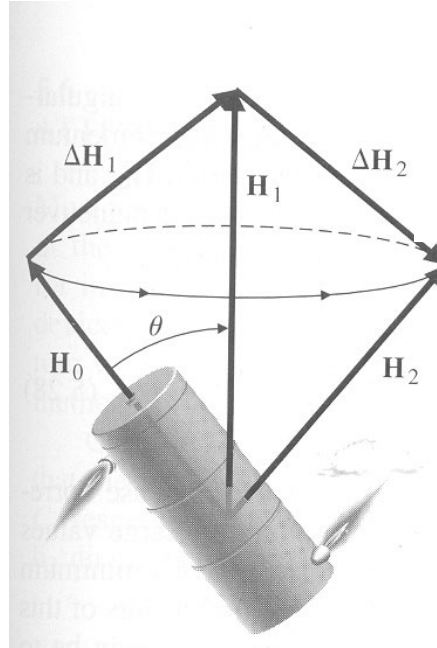
Thus,

$$\begin{aligned} (C\omega_f)^2 &= (C\omega_3)^2 + (A\omega_t)^2 \\ \omega_f &= \left[(\omega_3)^2 + \left(\frac{A}{C} \omega_t \right)^2 \right]^{1/2} \\ &> \omega_3 \end{aligned}$$

which show that the final spin axis angular velocity has increased due to nutation damping.

Spacecraft Reorientation

When spinning satellites execute a Homann transfer to geosynchronous orbit, an inclination change is required. This necessitates changing the spinning axis of the spacecraft from original orientation to the orientation necessary the second maneuver.



Assume the spacecraft is initially in a state of pure spin about symmetry axis and that it to be returned to a state of pure spin about this axis. Then, the inertial direction of the symmetry axis is to be changed in an angle in 2θ . This reorientation maneuver is performed by deliberately exciting the natural coning motion of the spacecraft and allowing it to precess through on half cycle.

An impulse torque $\vec{T}$ due to thruster

$$\vec{T} = \frac{\Delta \vec{H}}{\Delta t}$$

During this impulsive thruster burn, the spacecraft angular-momentum vector becomes

$$\vec{H}_1 = \vec{H}_0 + \Delta \vec{H}_1$$

Since the torque is perpendicular to the spin axis, the spacecraft spin rate $\dot{\phi}$ will be altered. Once the thruster placement is known, the torque supplied by the thruster can be calculated.

Since we also know the required change in the angular momentum, the thrusters must fire for a time interval of $\Delta t = \frac{|\Delta \vec{H}_1|}{|\vec{T}|}$.

The spacecraft now begins to precess at a rate given by

$$\dot{\psi} = \frac{C\dot{\phi}}{(A-C)\cos\theta} \quad \begin{array}{l} \dot{\phi}; \text{spin rate} \\ \dot{\psi}; \text{precession rate} \end{array}$$

In one-half precession cycle the spacecraft spin axis will reach the desired new position.

This will occur when ψ changes by π radian or it will occur a time interval as following after the original impulse.

$$\begin{aligned} \frac{\Delta\psi}{\Delta t} &= \frac{C\dot{\phi}}{(A-C)\cos\theta} \\ \rightarrow \Delta t &= \frac{(A-C)\cos\theta\Delta\psi}{C\dot{\phi}} = \frac{\pi(A-C)\cos\theta}{C\dot{\phi}} \end{aligned}$$

If the original spin rate is small, this will be a long time interval.

At this point, a second impulse, identical to first in magnitude, and with the same orientation with respect to the spacecraft, will cause the $\vec{H}_1$ to change by the almost of $\Delta\vec{H}_2$. The $|\vec{H}_2|$ is identical to the $|\vec{H}_0|$ and is once again aligned with the spacecraft spin axis.

So, after this second maneuver spacecraft is once again in a state of pure spin again and no longer coning.

Total cost

$$\Delta H_{tot} = 2|\Delta \vec{H}| = 2|\vec{H}_o| \tan \theta$$

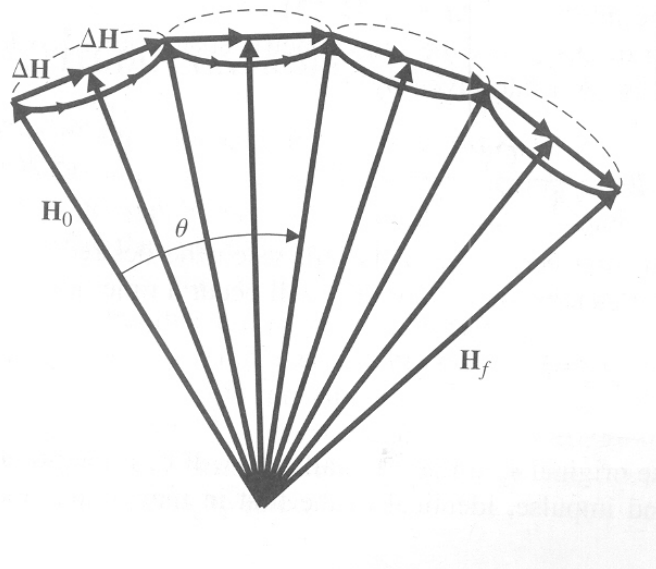
$$\tan \theta = \frac{|\Delta \vec{H}|}{|\vec{H}_o|}$$

This equation shows that reorientations by $\theta=90^\circ$ are infinitely expensive. Even moderately large value of θ can be quite costly.

A less expensive method to perform a large reorientation would then be to execute a series of small reorientation maneuvers.

If θ is small

$$\Delta H_{tot} = 2|\vec{H}_o|\theta$$



It is always cheaper to move a spinning spacecraft to a new orientation by a series of small coning maneuvers, rather than one large maneuver.

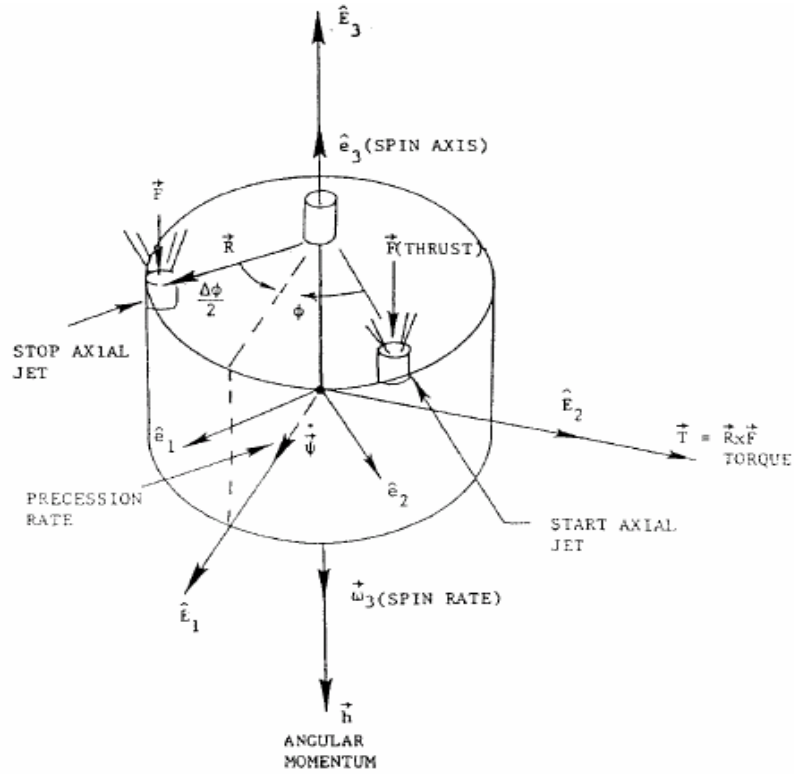


Figure 2.7 Pulsed-jet control for spin stabilized spacecraft.

$$T_{av} = \frac{2}{\Delta\phi} \int_0^{\frac{\Delta\phi}{2}} FR \cos \phi' d\phi' = \frac{2FR}{\Delta\phi} \sin \frac{\Delta\phi}{2}$$

$$\bar{T} = \frac{d\bar{H}}{dt} = \bar{R} \times \bar{F} \approx \frac{\Delta\bar{H}}{\Delta t}$$

$$\Delta H = H\theta \text{ per impulse}$$

$$\theta = \frac{\Delta H}{H} = \frac{T\Delta t}{H} = \frac{T_{av}\Delta t}{H}$$

The magnitude of angular momentum is the same before and after reorientation. The firing time (burn period)

$$\omega = \frac{\Delta\phi}{\Delta t} \rightarrow \Delta t = \frac{\Delta\phi}{\omega}, \quad \omega \text{ is spin rate}$$

The total reorientation angle per cycle: 2θ . The amount of propellant mass required to produce the total reorientation

$$\theta_{tot} = \frac{n}{2}(2\theta) \quad n : \text{number of thruster firing, 2 times per cycle}$$

$$W = \frac{\text{Total Impulse}}{I_{sp}} \Rightarrow \therefore m_p = \frac{nF\Delta t}{gI_{sp}}$$

F, R, θ are given $\rightarrow \Delta\phi, \Delta t, m_p$ can be calculated.